Project Proposal

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KV cache eviction is a popular field of research focusing on reducing the size of the key-value matrix used in standard LLMs, with the main goal of improving long-range dependency retrieval. We would like to explore potential new options for KV cache management and benchmark current best approaches. To do this, we can use open source models like Llama3 and easily replace their forward method with our own, optimized version. This is ideal for our scope, as it will not require any fine-tuning or training of a new model, simply running the inference step. Some of our preliminary ideas for KV cache management include using varying cache sizes for each layer, utilizing the attention weights to see what tokens are most "important", and potentially training a small model to learn what tokens might be needed later. We will likely refine and change our approach as we do more research on the current state of the art approaches and what has already been explored in depth.

Work Distribution

Gavin L, Max A: KV cache implementation
Max S: Testing
Beck S: Data visualization

Data Gathering Plan:

In order to compare our method against other techniques, we plan to use standardized open source benchmarks like needle in a haystack (NIAH) or RULER (Hsieh, 2024). These benchmarks are published on GitHub and compare models' ability to analyze and interpret long-range dependencies in input text. This, alongside memory size comparisons, will allow us to see if our model's memory footprint drops at the cost of inference ability.

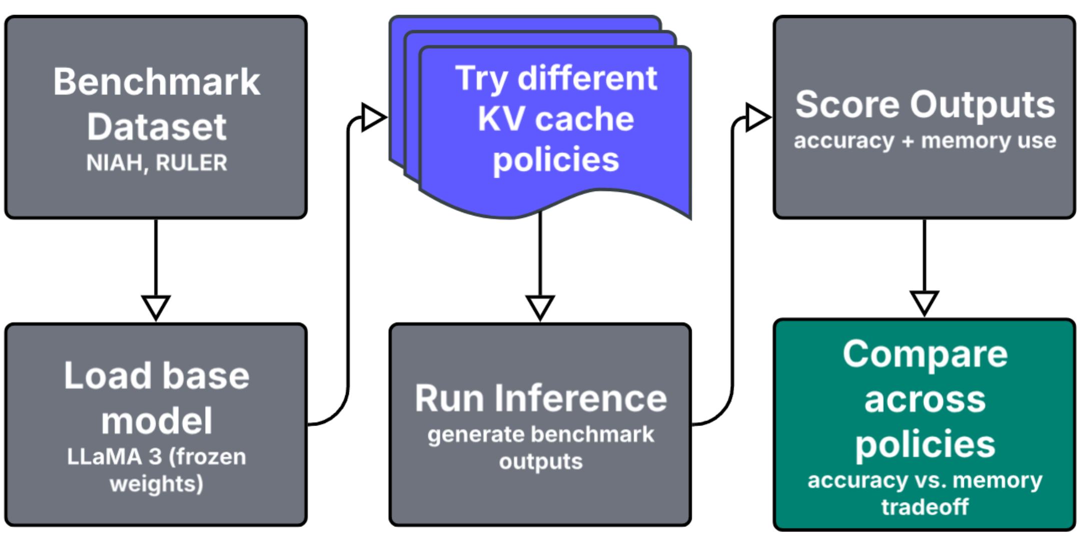

Figure 1: Comparative pipeline for evaluating KV cache policies on benchmark datasets

References

Hsieh, C.-P., Sun, S., Kriman, S., Acharya, S., Rekesh, D., Jia, F., Zhang, Y., & Ginsburg, B. (2024). RULER: What's the Real Context Size of Your Long-Context Language Models? arXiv.Org. https://arxiv.org/abs/2404.06654